AETHER EYE - SEMANTIC SLAM AND LIDAR MAPPING WITH MLLM GUIDED NAVIGATION FOR VISUAL ASSISTANCE

ABSTRACT

Our aim is to develop an assistive solution for visually impaired individuals that enables them to perform daily activities independently and safely, comparable to sighted individuals. To achieve this, the system is divided into indoor and outdoor navigation modules, each designed to address the specific challenges of its environment. Indoor navigation uses a camera feed integrated with ROS to capture and process environmental information. Outdoor navigation combines GPS data with camera input to understand the surrounding context. All sensor data are processed locally on a Raspberry Pi, where consecutive video frames are analysed to detect significant changes in the environment. When notable changes are identified, selected frames are forwarded to a vision language processing module, which analyses each frame and generates a conceptual textual description of the observed scene, event, or action. A memory based agent enhances this description by incorporating contextual information such as known individuals, familiar locations, or previously observed environments. The resulting text is converted into audio feedback using a text to speech engine, providing clear and intuitive guidance to the user.In addition to navigation support, an inertial measurement unit is integrated to enable fall detection and step counting, thereby improving user safety and activity monitoring. Together, these components form a unified system that delivers real time assistance, contextual awareness, and auditory guidance for visually impaired users.

We have put forward a proposal for our project below.

The Total Cost for the components is Rs 21978 . The details for each components have been listed below .

SL	COMPONENTS	PURCHASED COST
1	Raspberry Pi 4 (4 gb ram)	8142
2	LIDAR LUNA	2500
3	SD CARD	350
4	RASPBERRY PI CASE	500
5	PI CAM AUTOFOCUS IMX219	3836
6	ESP32	350
7	NEO 7M GPS	740
8	Connectors & Cables	500
9	Pro Range 11.1 V 1000 mah	1150
10	IMU	550
11	3D PRINTING DESIGN	3000
12	SD CARD READER	360
		21978

Proposed and verified by the below 5 team members :

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